



CMSC 491

Data Communication & Networking

Lec #4: Introduction to the Internet (IV)



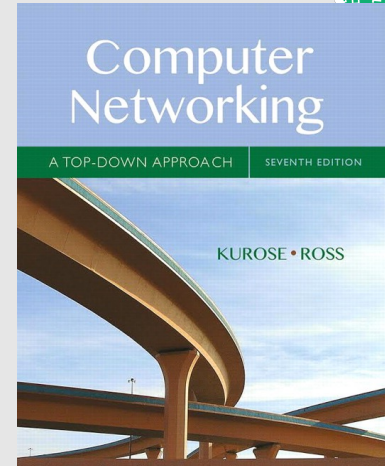
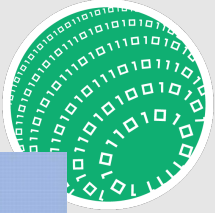
VCU

Tamer Nadeem

Introduction to the Internet

Overview

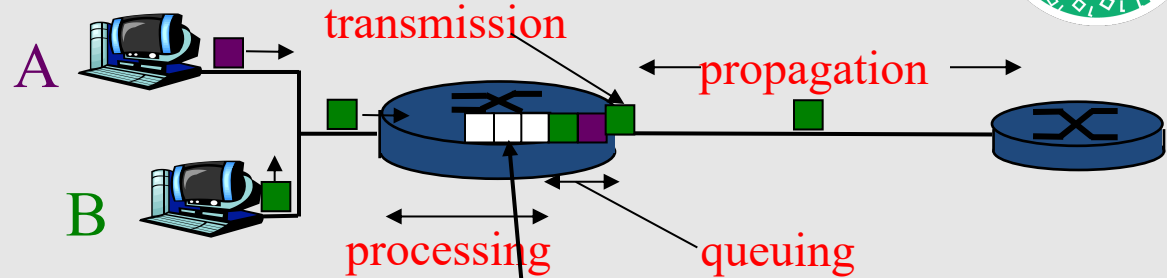
- What's the Internet (KR 1.1)
- Network Edge (KR 1.2)
- Network Core (KR 1.3)
- **Performance: Loss and Delay (KR 1.4) (cont'd)**
- Protocol Layering (KR 1.5)
- Networks Under Attack (KR 1.6)
- History (KR 1.7)



*Computer
Networking: A Top
Down Approach*
7th edition
Jim Kurose, Keith Ross
Pearson
May 2016

Performance: Loss and Delay

Delay in packet-switched networks



$$d_{hop} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

free (available) buffers: arriving packets dropped (lost) if no free buffers

– Processing Delay (d_{proc})

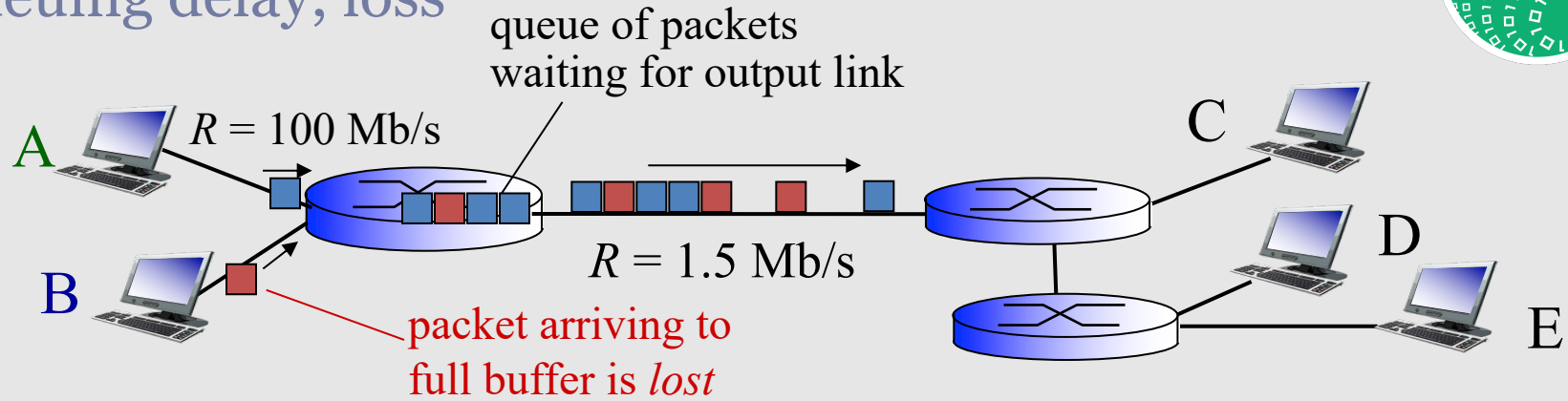
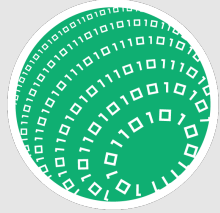
- Check for bit errors
- Determine the output interface to forward packet on

– Queuing Delay (d_{queue})

- Time spent waiting at outbound interface for transmission
- Duration depends on the level of congestion at the interface

Packet Switching

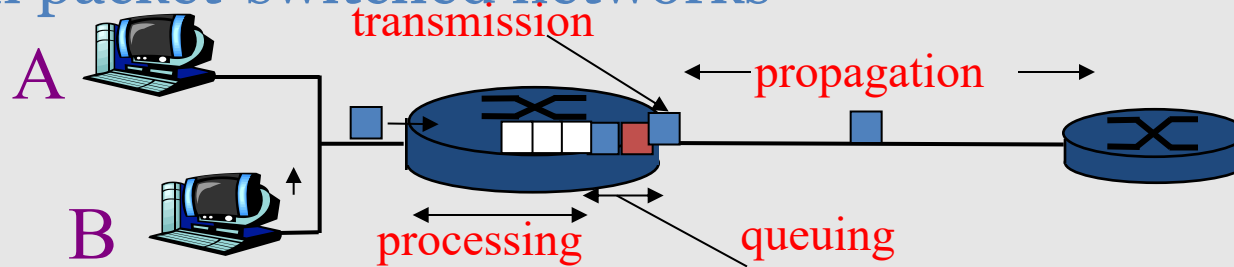
Queuing delay, loss



- If arrival rate (in bits) to link exceeds transmission rate (capacity) of link for a period of time:
 - packets will queue (buffered), wait to be transmitted on link
 - packets can be dropped (**lost**) if memory (**buffer**) fills up
 - Lost packet may be retransmitted by previous node, by source end system, or not at all

Performance: Loss and Delay

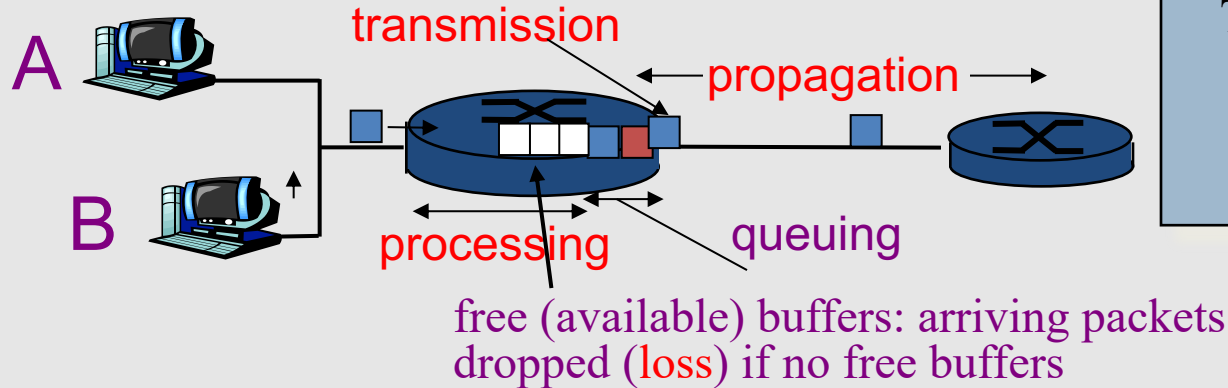
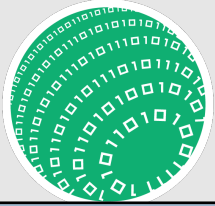
Delay in packet-switched networks



- **Typical transmission delay:**
 - L/R (significant for low-speed links)
 - $120 \mu\text{s}$ (for 1,500 byte packet on a 100 Mbps Ethernet)
- **Typical processing delay:**
 - a few micro-secs or less
- **Typical queuing delay:** Only delay that changes over time
 - depends on congestion: *packet arrival rate to link (temporarily) exceeds output link capacity*
- **Typical propagation delay:**
 - $\leq 1 \mu\text{s}$ on a small campus
 - $\approx 25 \text{ ms}$ to the West coast

Performance: Loss and Delay

Delay in packet-switched networks



Transmission delay = $120 \mu\text{s}$
Propagation delay = $10 \mu\text{s}$
Processing delay = $1 \mu\text{s}$
Queuing delay = $k \times 120 \mu\text{s}$

What is k ?

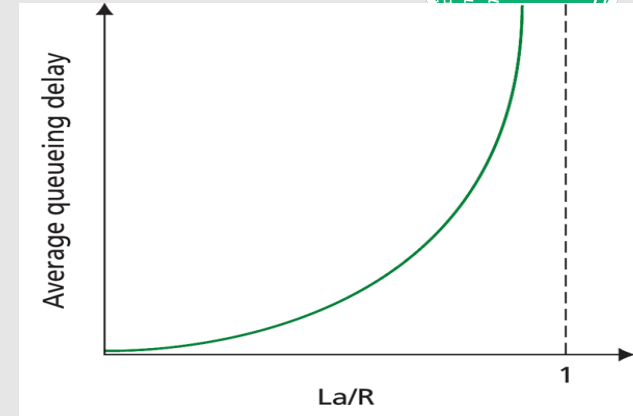
- ▶ What dominates end-to-end delay?
- ▶ Note that processing, transmission, and queuing delays are encountered at each hop and prop delay
 - ▶ End-to-end delay is largely a function of the number of routers encountered along the path from source to destination

Delay in Packet-Switched Networks

Queuing Delay

- What affects queuing delay?
 - traffic arrival rate (La)
 - Packet length in bits (L), average packet arrival rate (a)
 - speed of outgoing link (R)
 - nature of arriving traffic (uniform or bursty)
 - hard to quantify
- Represent queuing delay with statistical measures
 - average queuing delay
 - variance of queuing delay
 - Probability queuing delay exceeds some value
- Traffic intensity (La/R)
 - $La/R \sim 0$ - avg queuing delay small
 - $La/R \rightarrow 1$ - avg queuing delay large
 - $La/R > 1$ - more "work" arriving than can be serviced, avg delay infinite!

Having an arrival rate close to transmission rate is not safe

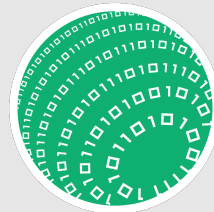


$La/R \sim 0$

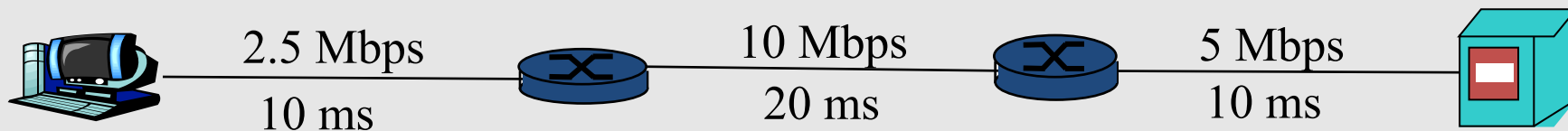
$La/R \rightarrow 1$

See Java applet: https://media.pearsoncmg.com/aw/ecs_kurose_compnetwork_7/cw/content/interactiveanimations/queuing-loss-applet/index.html

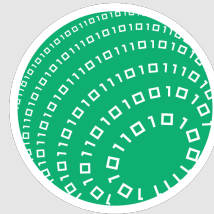
Question



- How long would it take a 5000-bit packet to travel from source to destination over the following network?

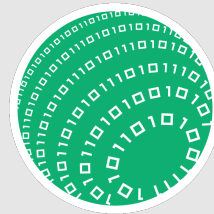


Questions



- What is the transmission delay for a 2000-byte packet over a 1 Mbps link?
- Where will be the last bit of the packet after the transmission delay has passed?
- What is the transmission delay for a 250,000-bit packet over a 1 Mbps link?

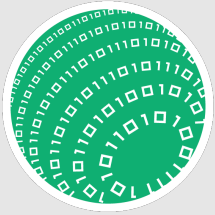
Questions



- What are the four types of delay that are (potentially) encountered at every router/hop? Which one may not be encountered?
- What is the difference between transmission delay and propagation delay?
- What two things affect the duration of queuing delay for a particular packet?

Performance: Loss and Delay

Example: What was the delay from my house?



- Pinging VCU from home

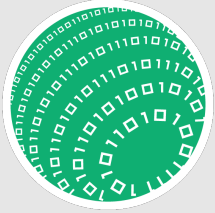
```
% ping www.vcu.edu
PING jupiter3.vcu.edu (128.172.22.56): 56 data bytes
64 bytes from 128.172.22.56: icmp_seq=0 ttl=245 time=16.315 ms
64 bytes from 128.172.22.56: icmp_seq=1 ttl=245 time=17.313 ms
64 bytes from 128.172.22.56: icmp_seq=2 ttl=245 time=17.228 ms
64 bytes from 128.172.22.56: icmp_seq=3 ttl=245 time=18.726 ms
^C
--- jupiter3.vcu.edu ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 16.315/17.395/18.726/0.862 ms
```

Varies based on queueing delay

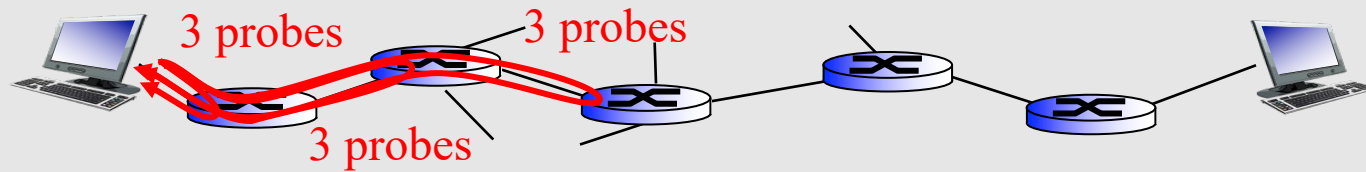
Round trip time: time between sending data and the response returning, equal to $2 * \text{end-to-end delay}$, and **roughly equal** $2 * \text{propagation delay}$ (assuming no queueing)

Performance: Loss and Delay

Traceroute



- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination.
 - traceroute from web: www.traceroute.org
- For all router i :
 - sends three packets that will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply



Performance: Loss and Delay



Example: `tracert` www.eurecom.fr //from `gaia.cs.umass.edu`

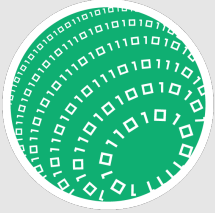
3 delay measurements from `gaia.cs.umass.edu` to `cs-gw.cs.umass.edu`

1	cs-gw (128.119.240.254)	1 ms	1 ms	2 ms
2	border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)	1 ms	1 ms	2 ms
3	cht-vbns.gw.umass.edu (128.119.3.130)	6 ms	5 ms	5 ms
4	jnl-at1-0-0-19.wor.vbns.net (204.147.132.129)	16 ms	11 ms	13 ms
5	jnl-so7-0-0-0.wae.vbns.net (204.147.136.136)	21 ms	18 ms	18 ms
6	abilene-vbns.abilene.ucaid.edu (198.32.11.9)	22 ms	18 ms	22 ms
7	nycm-wash.abilene.ucaid.edu (198.32.8.46)	22 ms	22 ms	22 ms
8	62.40.103.253 (62.40.103.253)	104 ms	109 ms	106 ms
9	de2-1.de1.de.geant.net (62.40.96.129)	109 ms	102 ms	104 ms
10	de.fr1.fr.geant.net (62.40.96.50)	113 ms	121 ms	114 ms
11	renater-gw.fr1.fr.geant.net (62.40.103.54)	112 ms	114 ms	112 ms
12	nio-n2.cssi.renater.fr (193.51.206.13)	111 ms	114 ms	116 ms
13	nice.cssi.renater.fr (195.220.98.102)	123 ms	125 ms	124 ms
14	r3t2-nice.cssi.renater.fr (195.220.98.110)	126 ms	126 ms	124 ms
15	eurecom-valbonne.r3t2.ft.net (193.48.50.54)	135 ms	128 ms	133 ms
16	194.214.211.25 (194.214.211.25)	126 ms	128 ms	126 ms
17	* * *			
18	* * *			
19	fantasia.eurecom.fr (193.55.113.142)	132 ms	128 ms	136 ms

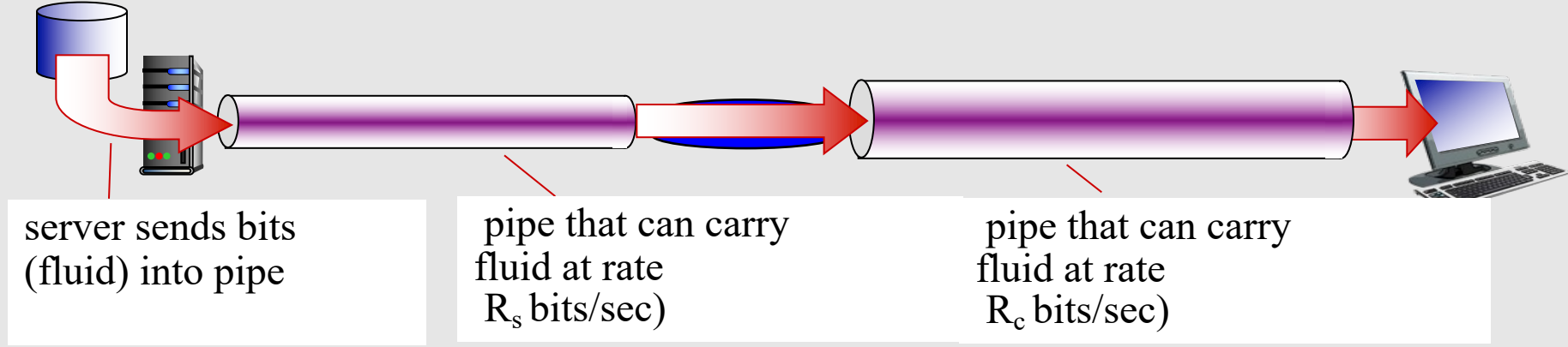
* means no response (probe lost, router not replying)

Performance: Loss and Delay

Throughput

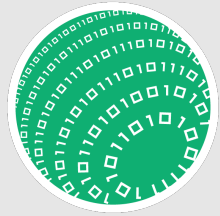


- *throughput*: rate (bits/time unit) at which bits transferred between sender/receiver
 - *instantaneous*: rate at given point in time
 - *average*: rate over long(er) period of time

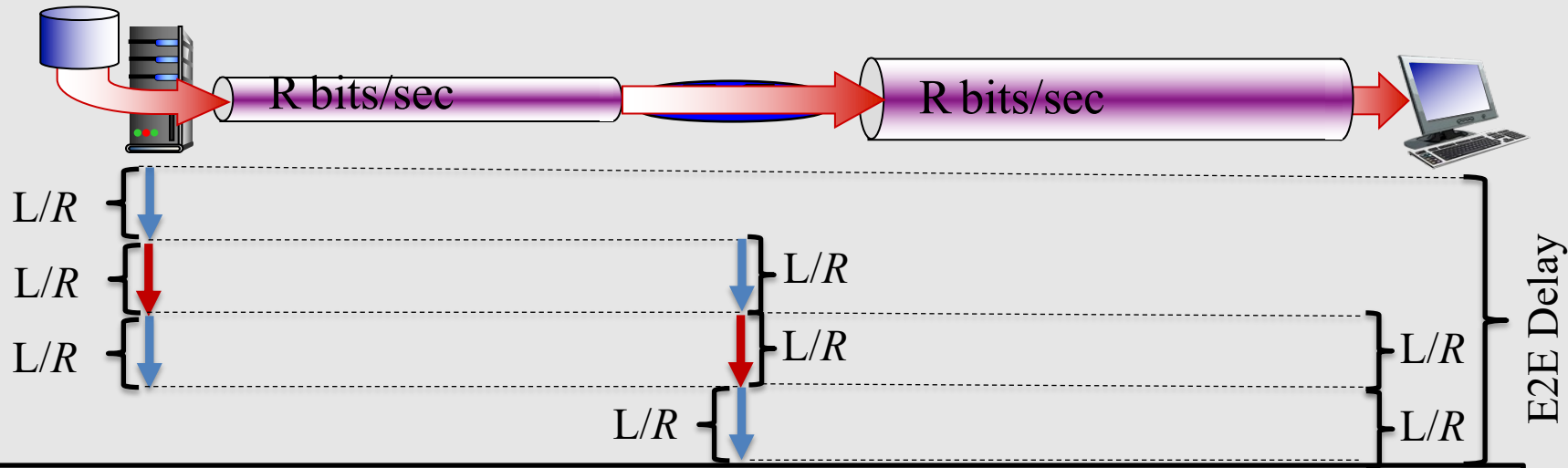


Performance: Loss and Delay

Throughput



- ▶ $R_s = R_c = R$ What is average end-end throughput?



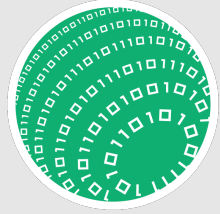
End-2-End Delay (N packets) = $2 \cdot L/R + (N-1) \cdot L/R = (N+1) \cdot L/R$ sec

Throughput (N packets) = transmitted bits / e2e delay

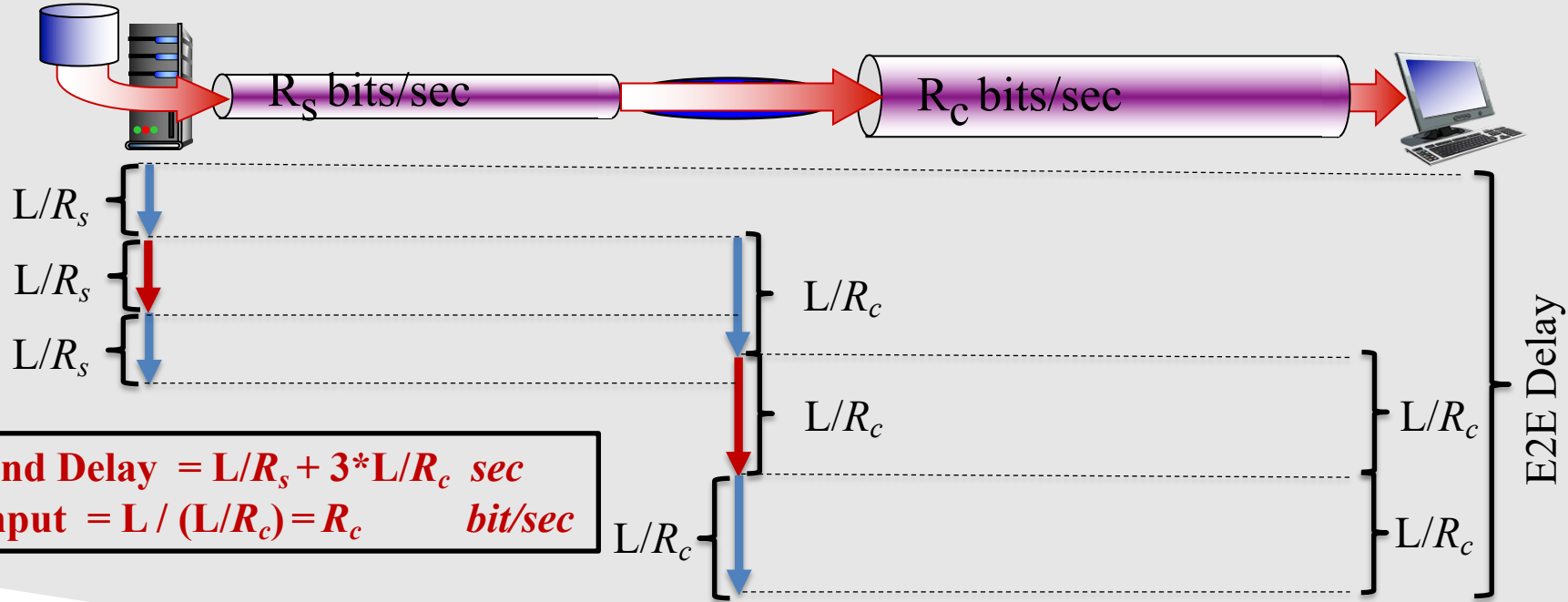
$$\begin{aligned}
 &= \frac{N \cdot L}{(N+1) \cdot L/R} = \frac{NR}{N+1} \text{ bit/sec} \\
 &\approx R \text{ bit/sec (for large } N) \\
 &= L / (L/R)
 \end{aligned}$$

Performance: Loss and Delay

Throughput



- ▶ $R_s > R_c$ What is average end-end throughput?

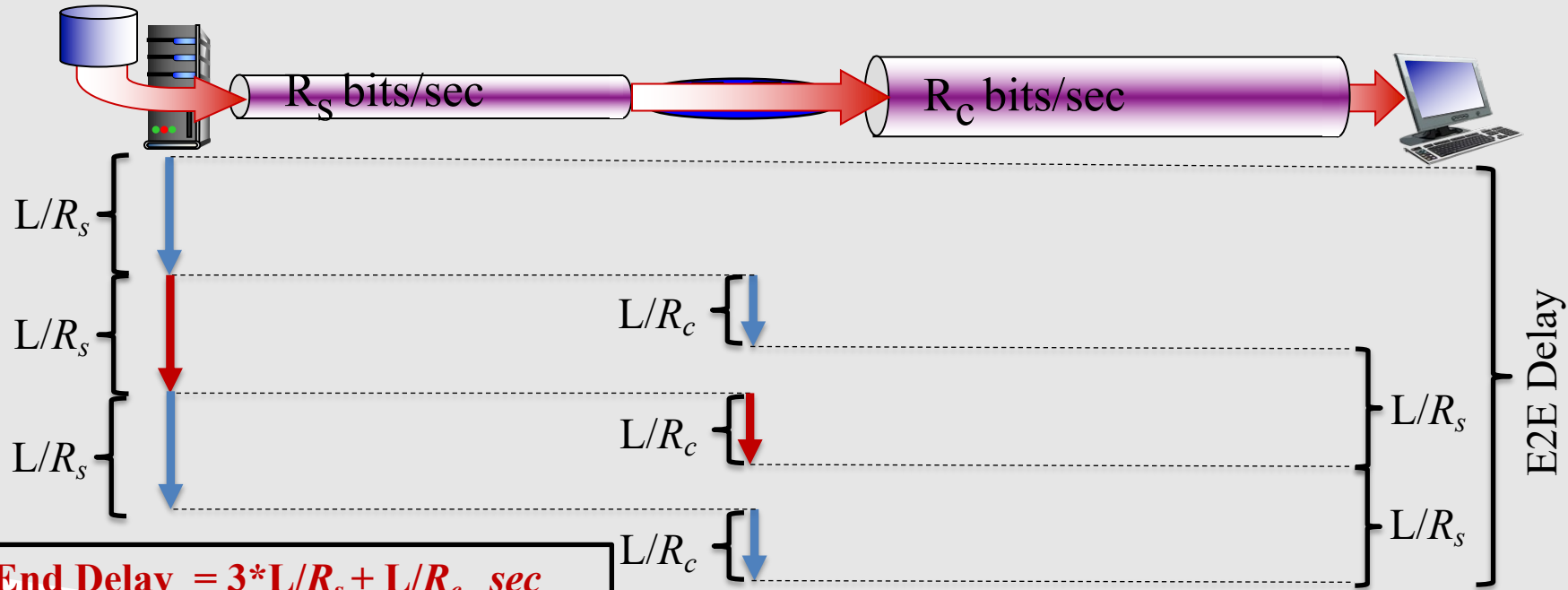


Performance: Loss and Delay

Throughput



- ▶ $R_s < R_c$ What is average end-end throughput?



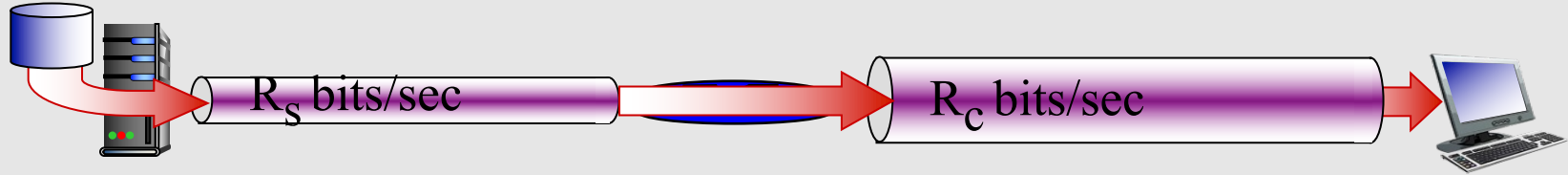
End-2-End Delay = $3 \cdot L/R_s + L/R_c$ sec
Throughput = $L / (L/R_s) = R_s$ bit/sec

Performance: Loss and Delay

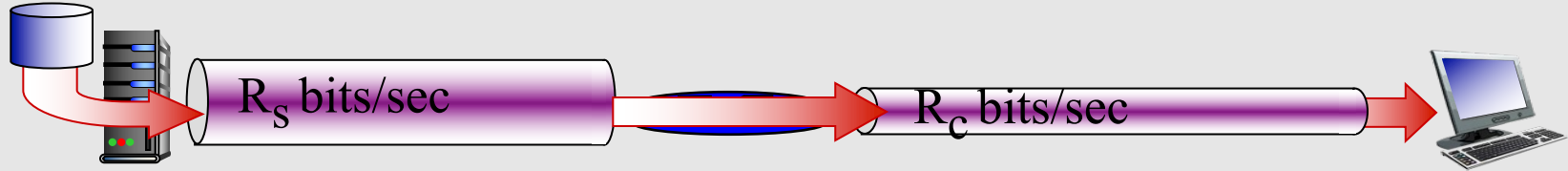
Throughput



- ▶ $R_s < R_c$ What is average end-end throughput?



- ❖ $R_s > R_c$ What is average end-end throughput?



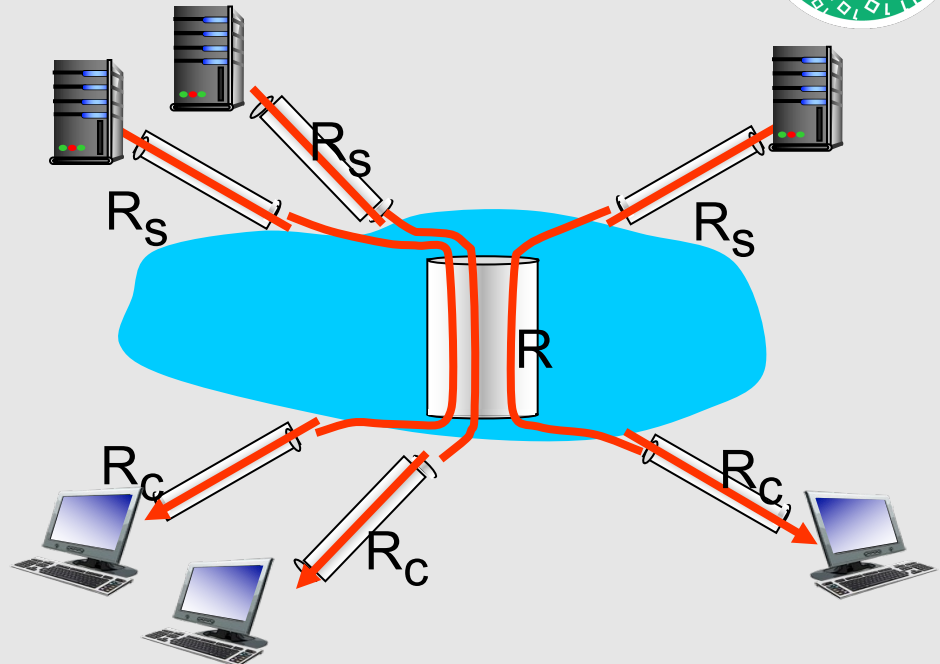
bottleneck link

link on end-end path that constrains end-end throughput

Performance: Loss and Delay

Throughput: Internet Scenario

- Per-connection end-to-end throughput:
 - $\min(R_c, R_s, R/10)$
- In practice:
 - R_c or R_s is often bottleneck

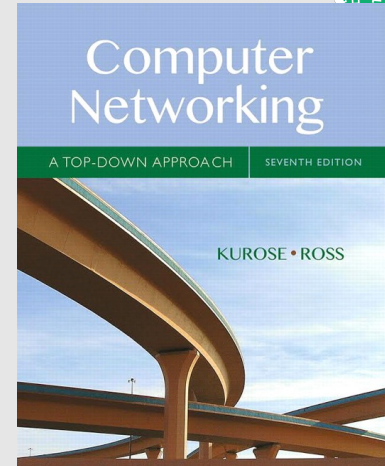


10 connections (fairly) share backbone
bottleneck link R bits/sec

Introduction to the Internet

Overview

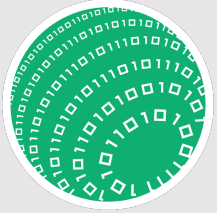
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Protocol Layering in the Internet

Why layering



*Networks are complex,
with many “pieces”:*

- ▶ hosts
- ▶ routers
- ▶ links of various media
- ▶ applications
- ▶ protocols
- ▶ hardware, software

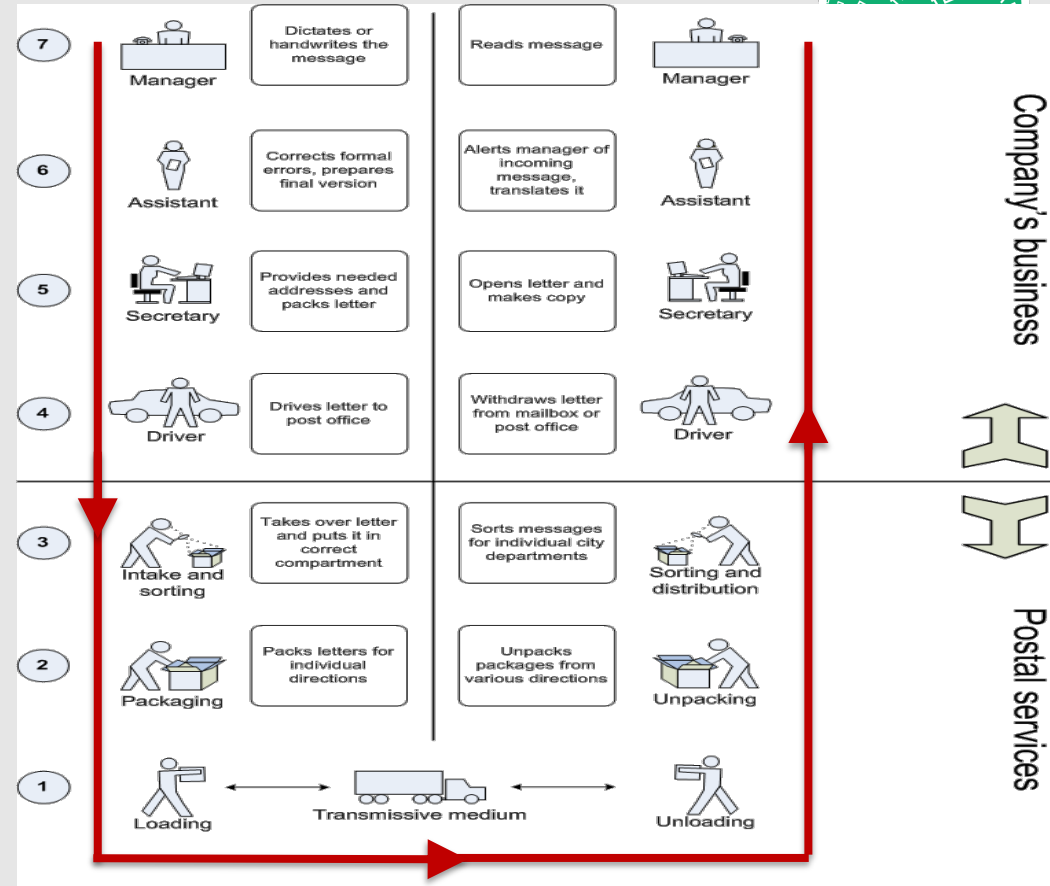
Dealing with complex systems:

- ▶ explicit structure allows identification, relationship of complex system's pieces
- ▶ modularization eases maintenance, updating of system
 - ▶ change of implementation of layer's service is transparent to rest of system

Protocol “Layers”

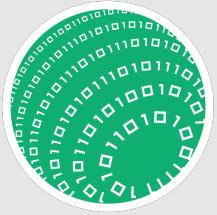
Layering of functionality

- **Layers:** each layer implements a service
 - via its own internal-layer actions
 - relying on services provided by layer below
- Layering considered harmful?

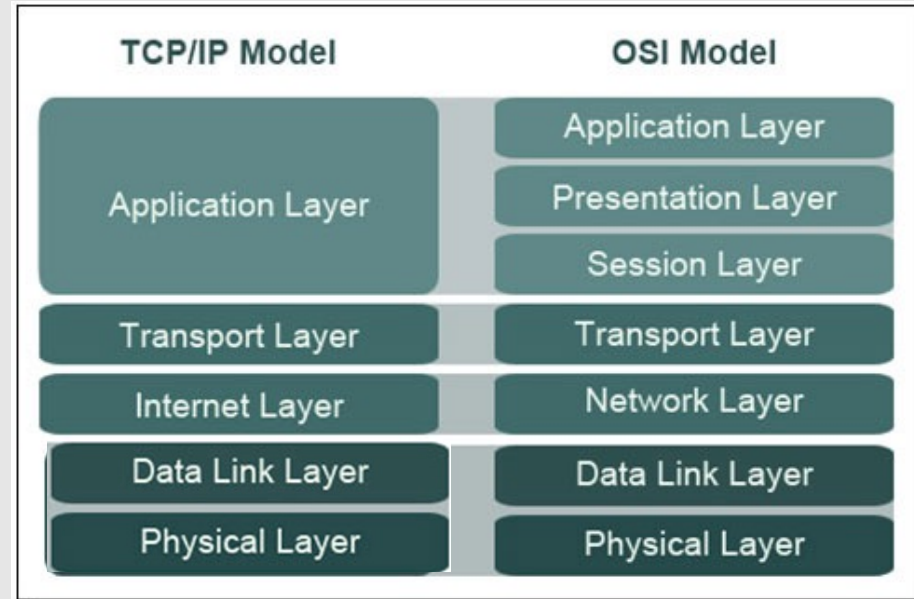


Protocol Layering in the Internet

ISO/OSI reference model



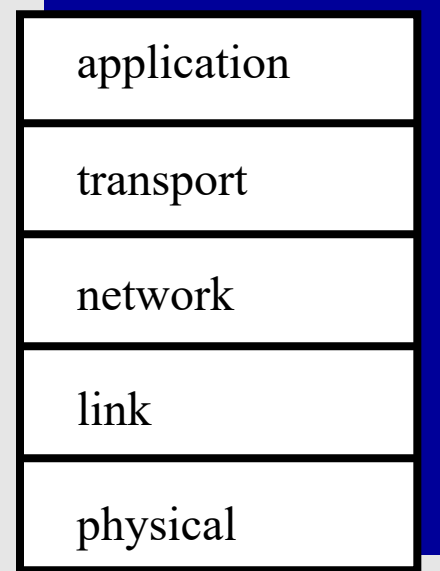
- ▶ ***presentation***: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ▶ ***session***: synchronization, checkpointing, recovery of data exchange
- ▶ Internet stack “missing” these layers!
 - ▶ these services, *if needed*, must be implemented in application
 - ▶ needed?



Protocol Layering in the Internet

Internet (TCP/IP) Model ("Stack")

- **Application layer**
 - Supporting network applications
 - FTP, SMTP, HTTP
- **Transport layer**
 - Host-host data transfer
 - TCP, UDP
- **Network layer**
 - Routing of packets from source to destination
 - IP, routing protocols
- **Link layer**
 - Data transfer between directly connected network elements
 - Ethernet, 802.11, SONET, ...
- **Physical layer**
 - The insertion of individual bits "on the wire"
 - Manchester encoding

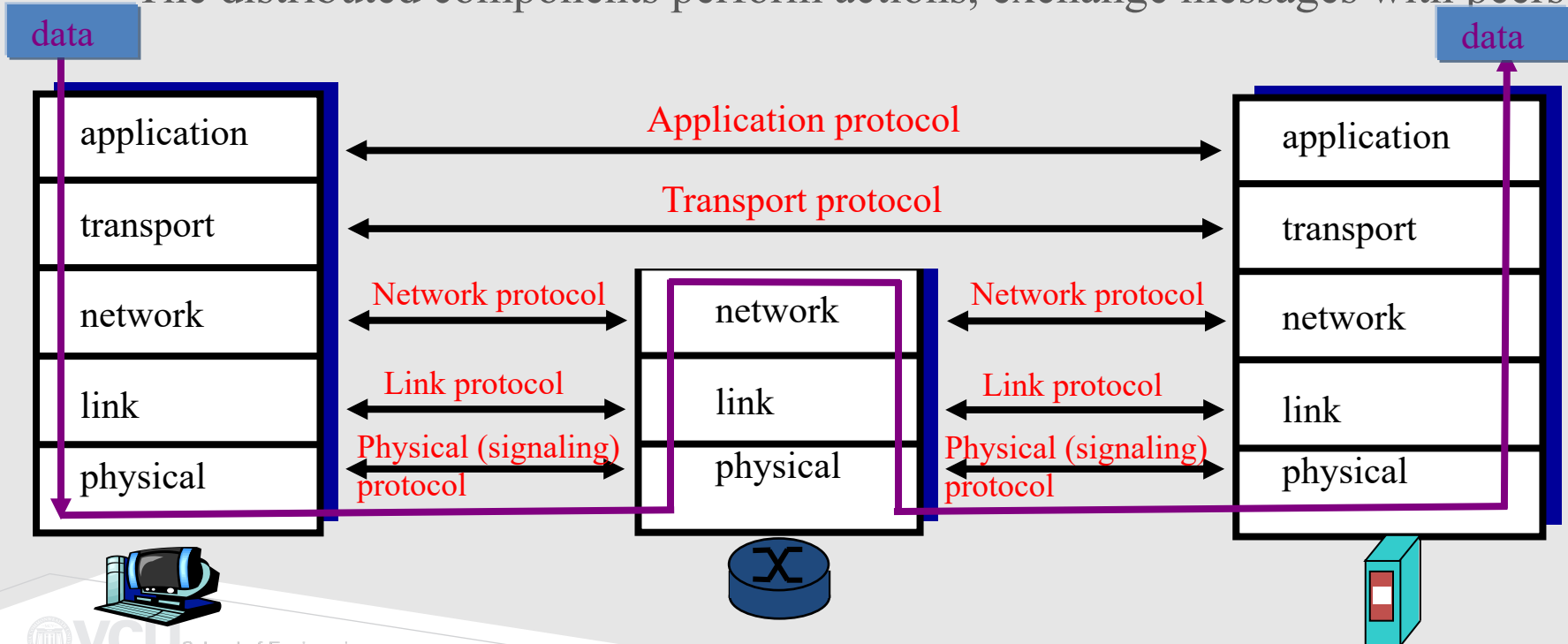


Different services
specified at each
layer interface

Protocol Layering in the Internet

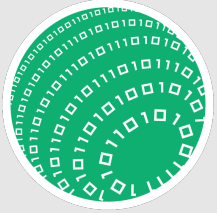
Internet protocol layers ("stack")

- Each layer implements a protocol with its peer layer in a distributed system
- The distributed components perform actions, exchange messages with peers

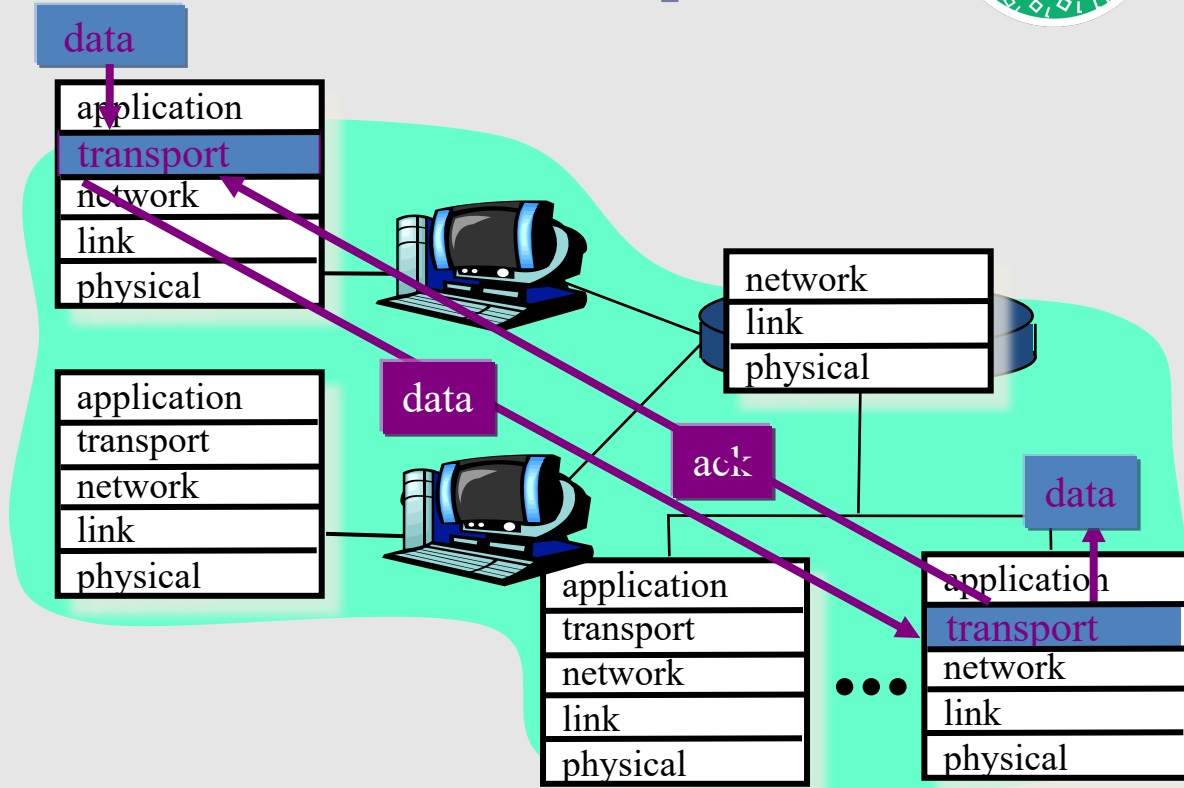


Protocol Layering in the Internet

Transport layer logical communication example



- Receive data from application
- Add transport-layer protocol information
- Send to peer transport layer
- Wait for peer transport layer to respond
- Peer transport delivers data to its application layer

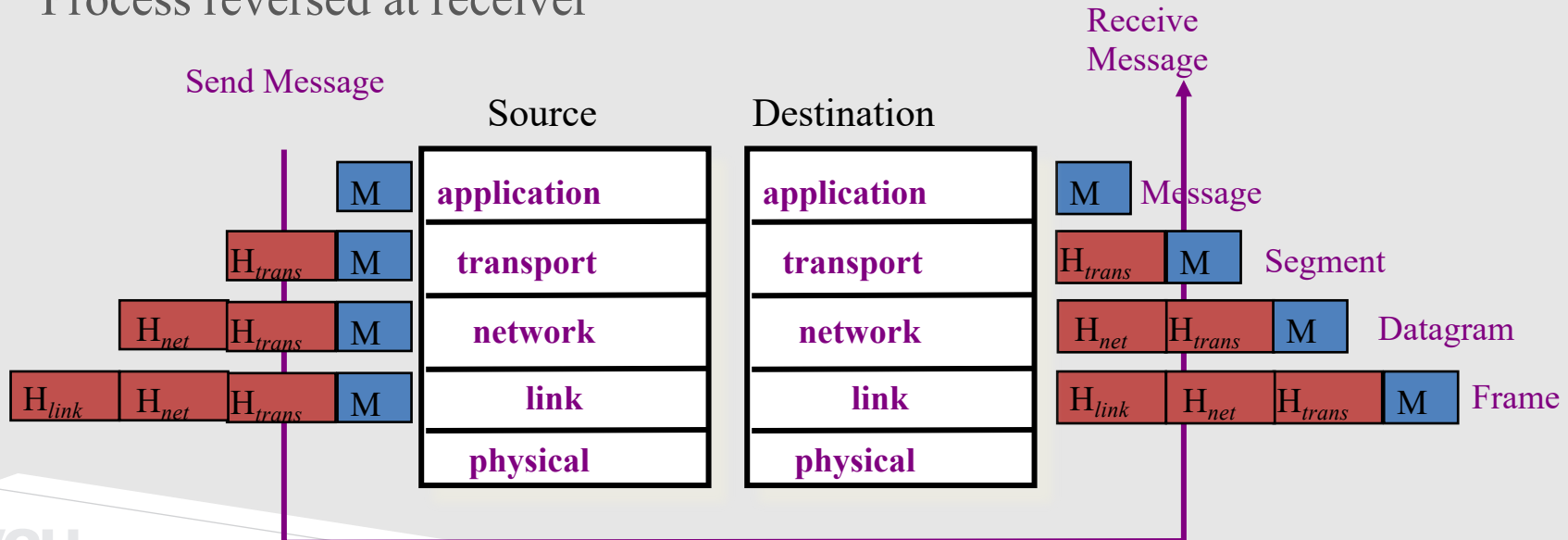


Protocol Layering in the Internet

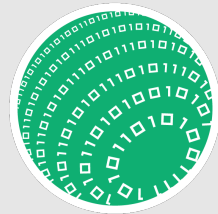
Protocol layering and data formats



- At sender, each layer takes data from above
 - Adds header information to create new data unit
 - Passes new data unit to layer below
- Process reversed at receiver



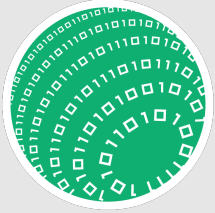
Questions



- What are the five network protocol layers?
- Which layers do end systems use?
- Which layers do routers use?

Introduction to the Internet

Summary



- Covered a "lot" of material
 - Internet overview
 - What's a protocol?
 - Network edge, core, access network
 - Performance: loss, delay
 - Layering and service models
- You now hopefully have:
 - Context, overview, "feel" of networking
 - More depth, detail later in course



Questions?

